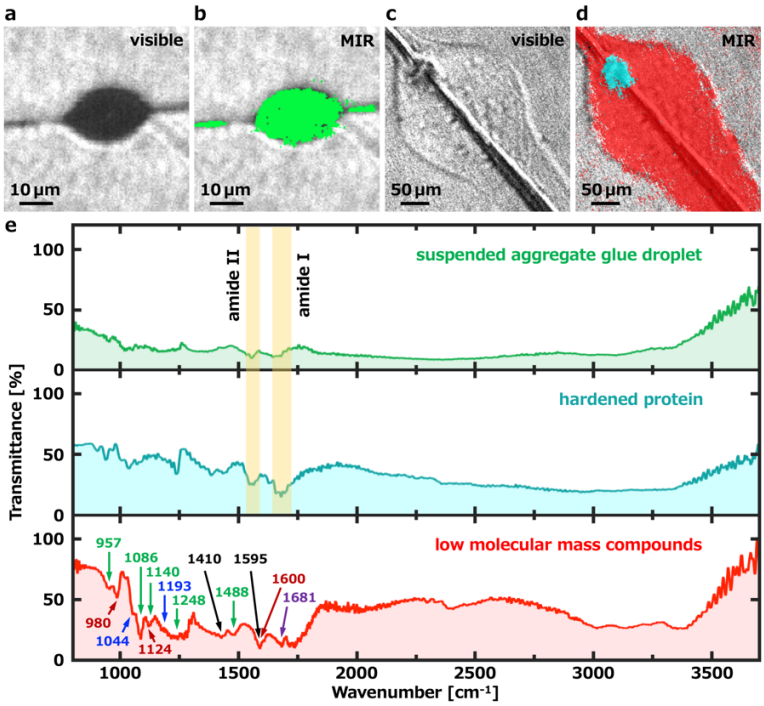

Figure 3 | MIR hyperspectral imaging and mapping of the aggregate glue droplet. Suspended aggregate glue droplet images of **a,** visible light illumination and **b,** mapped hyperspectral image. Attached aggregate glue droplet images of **c,** visible light illumination and **d,** mapped hyperspectral image. **e,** Infrared transmittance spectra of the suspended aggregate glue droplet, central and spread out area of attached aggregate glue droplet by extracted hyperspectral image. The mapped color is a pseudo-color.

Since low molecular mass compounds are a complex mixture, its spectrum is the sum of several kinds of substances. Except for choline dihydrogen phosphate, according to known constituents [10–16], the peaks at 1410 cm^{-1} ($\nu_s[COO^-]$) and 1595 cm^{-1} ($\nu_{as}[COO^-]$) are attributed to the carboxylate anion of γ-aminobutyramide, betaine and glycine [38], and the peak at 1681 cm^{-1} is attributed to the hydrophilic carbonyl group in the pyrrole ring of 2-pyrrolidone [39]. The presence of the sulfonate salt is seen by the O=S=O stretching vibrations at 1044 cm^{-1} ($\nu_{as}[O=S=O]$) and 1193 cm^{-1} ($\nu_s[O=S=O]$) [40,41]. The sulfonic acid functionality should be from *N*-acetyltaurine and isethionic acid. The NH_2 vibrations [42,43] at 980 cm^{-1} (NH_2 twisting), 1124 cm^{-1} (NH_2 wagging) and 1600 cm^{-1} (NH_2 scissoring) are attributed to amino group

of putrescine, *N*-acetylputrescine, *β*-alaninamide and *γ*-aminobutyramide. The hydrogen phosphate dianions can interact with putrescine to form a hydration system [43]. In aggregate glue droplets, putrescine dihydrogen phosphate may also perform dehydration and rehydration functions similar to choline dihydrogen phosphate.

4. Direct measurement of adhesion processes

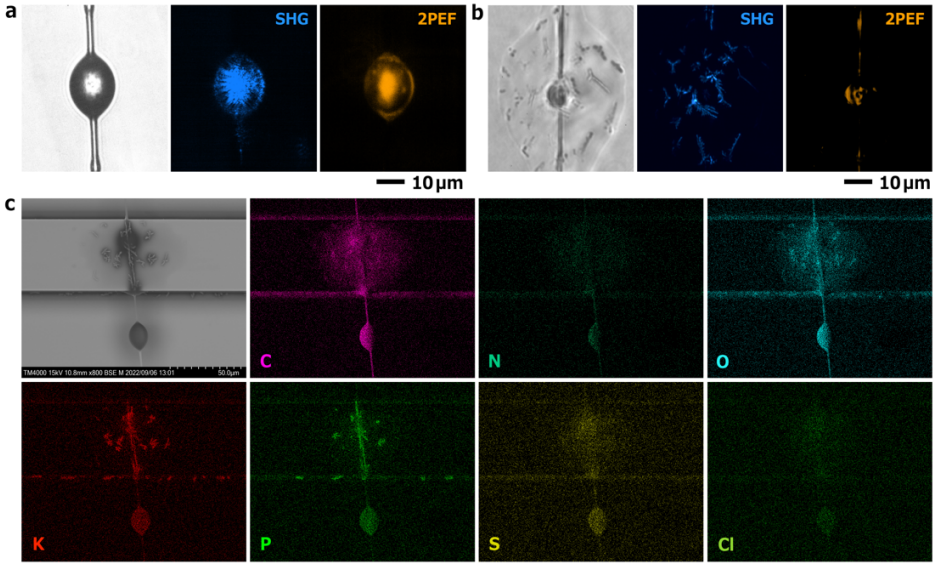

Figure 4 | Second harmonic generation (SHG), 2-photon excitation fluorescence (2PEF) imaging and scanning electron microscope energy dispersive spectrometry (SEM-EDS) mapping of the aggregate glue droplet. a, Suspended aggregate glue droplet microscopy images of white light illumination, SHG, and 2PEF. **b,** Attached aggregate glue droplet microscopy images of white light illumination, SHG, and 2PEF. **c,** SEM-EDS mapping images of the suspended and attached aggregate glue droplets. The color of SHG, 2PEF and SEM-EDS is a pseudo-color.

To investigate the behavior of aggregate glue droplets before and after adhesion, we performed selective observations of molecular structure and elemental mapping using second harmonic generation microscopy, 2-photon excitation fluorescence microscopy, and scanning electron microscope energy

dispersive spectrometry. Figures 4a and 4b show microscopic images of second harmonic generation and 2-photon excitation fluorescence microscopy in suspended and attached aggregate glue droplets, respectively. Although the second harmonic generation, 2-photon excitation fluorescence in Fig. 4a, and the fluorescence signals in Fig. 1b are all scattered light, the distribution of the signals is different in the images due to different excitation mechanisms. Since the suspended aggregate glue droplet approximates a sphere, the second harmonic generation and 2-photon excitation fluorescence signals show a relatively strong image in the middle. Based on the shape of the suspended aggregate glue droplet, the substances producing second harmonic generation and 2-photon excitation fluorescence signals should be uniformly distributed. On the other hand, since the attached aggregate glue droplet approximates a thin film, the signal distributions of the 2-photon excitation fluorescence in Figs. 4b and 1e are consistent, *i.e.*, the fluorescent signal originates from the protein. Second-harmonic microscopy can selectively observe structures with broken inversion symmetry. The originally uniform second harmonic generation and 2-photon excitation fluorescence signals exhibited local intensity distributions after the suspended aggregate glue droplet was attached to the CaF_2 substrate. Previous reports have observed second harmonic generation and 2-photon excitation fluorescence signals in suspended polymerized gel droplets, but the origin of the nonlinear optical response is unclear [44]. The branching structure presented by the second harmonic generation signal in Fig. 4b suggests that the origin of the second harmonic generation is a crystal structure. Subsequently, the elements in the aggregate glue droplet were analyzed and mapped by scanning electron microscope energy dispersive spectrometry. Here, the aggregate glue droplet is attached to a grooved Si substrate. As shown in Fig. 4c, suspended and attached aggregate glue droplets are observed simultaneously using the step of the substrate and groove. In the element mapping images, either suspended or attached aggregate glue droplet, C, N, S, Cl elements did not show local distribution, but in the attached area, the intensity distribution of K, P, O elements showed a branching structure similar to